MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 6

1. *Determine a basis for each of the following subspaces:*

(i) $H = \left\{ \begin{bmatrix} 4s \\ -3s \\ -t \end{bmatrix} : s, t \in \mathbb{R} \right\}$.

Every vector in H is of the form

$$\begin{bmatrix} 4s \\ -3s \\ t \end{bmatrix} = s \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

where s, t range freely over $\mathbb{R}$. Thus

$$H = \text{span} \left\{ \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Since these two vectors are linearly independent (there are two of them and they are not multiples of each other), the set

$$\left\{ \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a basis of H.

(ii) $K = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \; : \; a - 3b + c = 0 \right\}$.

We observe that $K = \text{Nul}(A)$ where

$$A = \begin{bmatrix} 1 & -3 & 1 & 0 \end{bmatrix}.$$

Since A is already in RREF, we see that a is the pivot variable and b, c, d are the free variables. Thus we get as the general solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$,

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 3b - c \\ b \\ c \\ d \end{bmatrix} = b \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

where b, c, d range freely over $\mathbb{R}$. Thus,

$$K = \text{Nul}(A) = \text{span}\left(\begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right).$$

It is clear that these three vectors are linearly independent (this follows from a routine argument involving the bottom 3 rows only). It follows that

$$\left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

forms a basis for K.

[Note: if you transform A into reduced echelon form and write $\text{Nul}(A)$ as span of vectors whose coefficients are the free variables (as we did above) then those vectors always will be linearly independent (Why? in the set above, since the free variables were x_3 and x_4, you can make an argument that the vectors are linearly independence using the 3rd and 4th entries of the column vectors (in **bold** above)). So here there is no need to check linear independence, since this particular method guarantees that they are!]

2. *Find a basis for the following subspaces:*

(i)

$$W_3 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2} : a = d = 0 \right\}.$$

Since a and d are 0, the variables b and c are unrestricted. Thus $\dim(W_3) = 2$. A basis for W_3 is given by

$$\left\{ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}.$$

(ii)

$$W_4 = \left\{ p(t) \in \mathbb{P}_2 : \frac{d}{dt} p(t) = 0 \right\}.$$

(Recall that $\mathbb{P}_2$ is the vector space of polynomials of degree at most 2).

W_4 consists only of the constant polynomials, i.e.

$$W_2 = \{a_0 + a_1 t + a_2 t^2 : a_1 = a_2 = 0\}.$$

Therefore, the members of W_2 are determined is a_0, thus $\dim(W_4) = 1$ and a basis for W_4 is $\{1\}$.

3. *Let $M_{2\times 2}$ be the vector space of all 2×2 matrices. Consider the two bases*

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

and

$$\mathcal{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$$

and $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in M_{2\times 2}$. *What are* $\mathbf{A}_{\mathcal{B}}$ *and* $\mathbf{A}_{\mathcal{C}}$*?*

To compute $\mathbf{v}_{\mathcal{B}}$, we must write $\mathbf{v}$ as a linear combination of of the vectors from $\mathcal{B}$:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

and we conclude that

$$\mathbf{v}_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

To compute $\mathbf{v}_{\mathcal{C}}$, we must write $\mathbf{v}$ as a linear combination of of the vectors from $\mathcal{C}$:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \frac{5}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{3}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

We conclude that

$$\mathbf{v}_{\mathcal{C}} = \begin{bmatrix} 5/2 \\ -3/2 \\ 5/2 \\ -1/2 \end{bmatrix}.$$

4. *Let A, B be two 4×3 matrices. Let $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ be the columns of A and let $\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3$ be the columns of B.*

(i) Suppose that $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are linearly independent. Find a basis for $\mathrm{Col}(A)$ and a basis for $\mathrm{Nul}(A)$. What are the dimensions of $\mathrm{Col}(A)$ and $\mathrm{Nul}(A)$?

$\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ is a basis of $\mathrm{Col}(A)$, because it spans $\mathrm{Col}(A)$ (they are the columns of A!) and they are linearly independent (by assumption). Since the basis of $\mathrm{Col}(A)$ has size 3, the dimension of $\mathrm{Col}(A)$ is 3. Since the rank of A is 3 which is the total number of columns of A, the homogeneous equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution (since there are no free variables). Hence

$$\mathrm{Nul}(A) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} = \mathrm{span}(),$$

i.e., it is the span of the set of no vectors. Technically speaking in this case we say that the empty set $\emptyset$ is a basis for $\mathrm{Nul}(A)$, and the dimension of $\mathrm{Nul}(A)$ is zero since the number of vectors in its basis is zero.

(ii) Suppose that $\mathbf{b}_1, \mathbf{b}_2$ are linearly independent and $\mathbf{b}_3 = 2\mathbf{b}_1 + 7\mathbf{b}_2$. Find a basis for $\mathrm{Col}(B)$ and a basis for $\mathrm{Nul}(B)$.

$\{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis of $\mathrm{Col}(B)$: by assumption it is linearly independent and $\mathbf{b}_3 = 2\mathbf{b}_1 + 7\mathbf{b}_2$, so we can show each column of B as a linear combination of $\mathbf{b}_1$ and $\mathbf{b}_2$. Hence, we have $\mathrm{span}(\mathbf{b}_1, \mathbf{b}_2) = \mathrm{span}(\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3) = \mathrm{Col}(B)$. Also $\dim \mathrm{Col}(B) = 2$ since the basis $\{\mathbf{b}_1, \mathbf{b}_2\}$ contains 2 vectors.
Furthermore, we know that $\dim \mathrm{Nul}(B) = 3 - \dim \mathrm{Col}(B) = 3 - 2 = 1$, so it suffices to find a single nonzero vector in $\mathrm{Nul}(B)$. Note that by assumption,

$$B \begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix} = 2\mathbf{b}_1 + 7\mathbf{b}_2 - \mathbf{b}_3 = \mathbf{0}$$

which shows that

$$\begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix} \in \mathrm{Nul}(B).$$

Thus

$$\mathrm{Nul}(B) = \mathrm{Span} \left\{ \begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix} \right\}$$

and a basis for $\mathrm{Nul}(B)$ is

$$\left\{ \begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix} \right\}.$$

5. *True or False? Give arguments why! In the questions below, V is an arbitrary vector space and A is an arbitrary $m \times n$ matrix.*

(1) The number of pivot columns of a matrix equals the dimension of its column space.

True.

(2) A plane in $\mathbb{R}^3$ is a two-dimensional subspace of $\mathbb{R}^3$.

False. A plane in $\mathbb{R}^3$ might not even be a subspace (it will be a subspace iff it contains the origin). If a plane in $\mathbb{R}^3$ does contain the origin, then it will be a two-dimensional subspace.

(3) The dimension of the vector space $\mathbb{P}_4$ (the vector space of polynomials of degree at most 4) is 4.

False. $\mathbb{P}_4$ has dimension 5. A basis for $\mathbb{P}_4$ is $\{1, t, t^2, t^3, t^4\}$ which contains five elements.

(4) If $\dim V = n$ and S is a linearly independent set in V, then S is a basis for V.

False. S could contain fewer than n elements, thus not spanning all of V. For example, $V = \mathbb{R}^3$ and $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.

(5) If a set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ spans a vector space V and if T is a set of more than p vectors in V, then T is linearly dependent.

True.

(6) If $\dim V = n$ and if S spans V, then S is a basis of V.

False. S could have more than n vectors. For example, if $V = \mathbb{R}^2$, $n = 2$ and $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$. S is not a basis for $\mathbb{R}^2$.

(7) If B is an Echelon Form of A, and if B has three nonzero rows, then the first three rows of A form a basis for the row space $\mathrm{Col}(A^T)$ of A.

False. For instance consider

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Any Echelon form of A will have three nonzero rows. The row space of A is all of $\mathbb{R}^3$, but the (transposes of the) first three rows of A will not be a basis for $\mathbb{R}^3$.

(8) The dimensions of the row space and the column space of A are always the same, even if A is not a square matrix.

True.

(9) The sum of the dimensions of the row space and the null space of A equals the number of rows in A.

False. Consider $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$ We have $\dim \mathrm{Col}(A^T) + \dim \mathrm{Nul}(A) = 1 + 1 = 2$, whereas A has only 1 row.

(10) If B is an echelon form of A, then the pivot columns of B form a basis for the column space of A.

False. Consider $A = \begin{bmatrix}1\\1\end{bmatrix}$ and its RREF $B = \begin{bmatrix}1\\0\end{bmatrix}$. Clearly span $\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\} \neq$ span $\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$

(11) The dimension of the null space of A is the number of columns of A that are *not* pivot columns.

True. This is a result of the course.

(12) If two matrices A and B are row equivalent, then the vector spaces $\mathrm{Col}(A)$ and $\mathrm{Col}(B)$ have the same dimension.

True. If they are row equivalent, then the same the same echelon form. Then count the number of pivots and the result follows.

6. (1) *Construct a* 2×2*-matrix whose nullspace equals its column space.*

$\begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}$

(2) Why does no 3×3-matrix have a nullspace that equals its column space?

Note that $\dim \mathrm{Col}(A) + \dim \mathrm{Nul}(A) = 3$. If $\mathrm{Nul}(A) = \mathrm{Col}(A)$, in particular, they should also have the same dimension, which is impossible.

7. *Construct a matrix with the required property, or explain why such a matrix cannot exist.*

(1) Column space contains $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$, row space contains $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\5\end{bmatrix}$.

$A = \begin{bmatrix}1 & 0\\1 & 0\\0 & 1\end{bmatrix}$. The column space claim is clear. Then note that the row space of A is all of $\mathbb{R}^2$.

(2) Column space has basis $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$, nullspace has basis $\left\{\begin{bmatrix}3\\2\\1\end{bmatrix}\right\}$.

No such matrix can exist. All such matrices must be 3×3. As such, we would need $\dim \mathrm{Col}(A) + \dim \mathrm{Nul}(A) = 3$.

(3) Dimension of nullspace = 1 + dimension of left nullspace.

$A = \begin{bmatrix}1 & 0\end{bmatrix}$. In this case, $\mathrm{rank}(A) = \dim \mathrm{Col}(A) = \dim \mathrm{Col}(A^T) = 1$. Thus $\dim \mathrm{Nul}(A) = 2 - 1 = 1$ and $\dim \mathrm{Nul}(A^T) = 1 - 1 = 0$. Actually, any matrix of size $m \times m + 1$ would work (where $m \geq 1$).

(4) Left nullspace contains $\begin{bmatrix}1\\3\end{bmatrix}$, row space contains $\begin{bmatrix}3\\1\end{bmatrix}$.

$A = \begin{bmatrix} 3 & 1 \\ -1 & -1/3 \end{bmatrix}$. The top row ensures the row space requirement. Multiplying on the left by the row vector $\begin{bmatrix} 1 & 3 \end{bmatrix}$ ensures the left nullspace requirement.

8. *Let $\mathcal{A} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ be a basis of $\mathbb{R}^3$.*

(1) Find a 3×3-matrix B such that $\mathbf{v} = B\mathbf{v}_{\mathcal{A}}$ for all $\mathbf{v} \in \mathbb{R}^3$.

Since $\mathcal{A}$ is a basis of $\mathbb{R}^3$, there exists $c_1, c_2, c_3 \in \mathbb{R}$ such that $\mathbf{v} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3$. Then we also have $v_{\mathcal{A}} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$. By the definition of the matrix multiplication, we obtain

$$\mathbf{v} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} v_{\mathcal{A}},$$

i.e. $B = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix}$.

(2) Find a 3×3-matrix C such that $\mathbf{v}_{\mathcal{A}} = C\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^3$.

Since $\mathbf{v} = B\mathbf{v}_{\mathcal{A}}$ for all $\mathbf{v} \in \mathbb{R}^3$ and B is invertible, we immediately have $\mathrm{B}^{-1}\mathbf{v} = \mathbf{v}_{\mathcal{A}}$ for all $\mathbf{v} \in \mathbb{R}^3$, i.e., $C = B^{-1}$.

9. *Consider the following eight vectors in $\mathbb{R}^8$:*

$$\mathcal{B} = (\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}), \quad \boldsymbol{w} = \begin{bmatrix} 100 \\ 200 \\ 44 \\ 50 \\ 20 \\ 20 \\ 4 \\ 2 \end{bmatrix}.$$

(1) Show that $\mathcal{B}$ is a basis of $\mathbb{R}^8$.

One solution would be to count the number of pivots in the reduced row echelon form of the matrix whose columns are the members of $\mathcal{B}$. There will be pivots in each of the columns of this matrix, so $\mathcal{B}$ is linearly independent. As $\mathcal{B}$ has eight elements, this means it is a basis for $\mathbb{R}^8$.

We will show this is independent by inspection. Suppose we have $\mathcal{B} = \{w_1, \ldots, w_8\}$, and $a_1w_1 + \ldots + a_8w_8 = 0$. Note first that w_1 and w_5 the only ones to be nonzero in the first and second entry. Looking at these entries, we must have that $a_1 = a_2 = 0$. Looking at the third/forth entry and so on, we can show that all the a_i are 0, so $\mathcal{B}$ is a set of 8 linearly independent vectors in $\mathbb{R}^8$ and thus a basis.

Alternatively, note that the vectors in the collection $\mathcal{B}$ are non-zero and mutually orthogonal and hence linearly independent. Since $\dim(\mathbb{R}^8) = 8$ and $|\mathcal{B}| = 8$, we conclude $\mathcal{B}$ is a basis for $\mathbb{R}^8$.

(2) Calculate $\boldsymbol{w}_{\mathcal{B}}$.

We have to solve:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 100 \\ 200 \\ 44 \\ 50 \\ 20 \\ 20 \\ 4 \\ 2 \end{bmatrix}.$$

Solving this system, we see that

$$\boldsymbol{w}_{\mathcal{B}} = \begin{bmatrix} 150 \\ 47 \\ 20 \\ 3 \\ 50 \\ 3 \\ 0 \\ -1 \end{bmatrix}.$$

(3) Suppose Alice needs to inform Bob about $\boldsymbol{w}$. However, Alice can transmit only four non-zero numbers (and not eight) to Bob; that is Alice can only send Bob a vector in $\mathbb{R}^8$ which has at most four non-zero entries. Luckily Bob does not need to know the exact values in $\boldsymbol{w}$, a good approximation suffices. Using your result in (2), which vector should Alice communicate to Bob?

Alice should communicate only the values in $\boldsymbol{w}_{\mathcal{B}}$ that are far away from 0. Thus Alice should send

$$\boldsymbol{v}_{\mathcal{B}} := \begin{bmatrix} 150 \\ 47 \\ 20 \\ 0 \\ 50 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Bob would use $\boldsymbol{v}_{\mathcal{B}}$ to reconstruct a vector $\boldsymbol{v}$ whose coordinate vector with respect to $\mathcal{B}$ is $\boldsymbol{v}_{\mathcal{B}}$. So

$$\boldsymbol{v} = 150\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 47\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 20\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + 50\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 100 \\ 200 \\ 47 \\ 47 \\ 20 \\ 20 \\ 0 \\ 0 \end{bmatrix}.$$

That is not exactly $\boldsymbol{w}$, but it is close enough for Bob's purposes.

This is the idea behind many data compression methods, in particular JPEG. Note that the choice of the basis $\mathcal{B}$ is crucial and we used the fact that always two entries in $\boldsymbol{w}$ that are next to each other, are similar in value. If you use JPEG to compress an image, then JPEG will be more efficient if the image is rather uniform (ie big parts of it are of similar colour).

10. *Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation with*

$$T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

(1) Consider the basis $\mathcal{B}_1 = \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ of $\mathbb{R}^2$ and the basis $\mathcal{B}_2 = \left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$ of $\mathbb{R}^3$. Determine $T_{\mathcal{B}_2,\mathcal{B}_1}$.

(In these solutions, we use colors for scalars to help the reader keep track of them)

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = T\left(\frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}\right) = \frac{1}{2}T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) + \frac{1}{2}T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix}5\\0\\1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}\frac{5}{2}\\\frac{1}{2}\\\frac{1}{2}\end{bmatrix}$$

$$= \underbrace{\frac{5}{2}\begin{bmatrix}1\\0\\0\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0\\1\\0\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\text{lin. combo. of vectors from } \mathcal{B}_2}.$$

and

$$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = T\left(-\frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}\right) = -\frac{1}{2}T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) + \frac{1}{2}T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = -\frac{1}{2}\begin{bmatrix}5\\0\\1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}-\frac{5}{2}\\\frac{1}{2}\\-\frac{1}{2}\end{bmatrix}$$

$$= \underbrace{-\frac{5}{2}\begin{bmatrix}1\\0\\0\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0\\1\\0\end{bmatrix} - \frac{1}{2}\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\text{lin. combo. of vectors from } \mathcal{B}_2}.$$

Then

$$T_{\mathcal{B}_2\mathcal{B}_1} = \begin{bmatrix}\frac{5}{2} & -\frac{5}{2}\\\frac{1}{2} & \frac{1}{2}\\\frac{1}{2} & -\frac{1}{2}\end{bmatrix}.$$

(2) Is $T(\mathbf{v}) = T_{\mathcal{B}_2\mathcal{B}_1}\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^2$?

Yes, in this case $T(\mathbf{x}) = T_{\mathcal{B}_2,\mathcal{B}_1}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^2$ (since both $\mathcal{B}_1$ and $\mathcal{B}_2$ are standard basis). For example,

$$T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) = \begin{bmatrix}\frac{5}{2} & -\frac{5}{2}\\\frac{1}{2} & \frac{1}{2}\\\frac{1}{2} & -\frac{1}{2}\end{bmatrix}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}5\\0\\1\end{bmatrix}.$$

(3) Consider the basis $\mathcal{C}_1 := \left\{\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$ of $\mathbb{R}^2$ and the basis $\mathcal{C}_2 = \left\{\begin{bmatrix}5\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ of $\mathbb{R}^3$. Determine $T_{\mathcal{C}_2,\mathcal{C}_1}$.

For (ii),

$$T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) = \begin{bmatrix}5\\0\\1\end{bmatrix} = \underbrace{1\begin{bmatrix}5\\0\\1\end{bmatrix} + 0\begin{bmatrix}0\\1\\0\end{bmatrix} + 0\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\text{lin. combo. of vectors from } \mathcal{C}_2},$$

and

$$T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = \begin{bmatrix}0\\1\\0\end{bmatrix} = \underbrace{0\begin{bmatrix}5\\0\\1\end{bmatrix} + 1\begin{bmatrix}0\\1\\0\end{bmatrix} + 0\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\text{lin. combo. of vectors from } \mathcal{C}_2}.$$

Then

$$T_{\mathcal{C}_2\mathcal{C}_1} = \begin{bmatrix}1 & 0\\0 & 1\\0 & 0\end{bmatrix}.$$

(4) Is $T(\mathbf{v}) = T_{\mathcal{C}_2,\mathcal{C}_1}\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^2$?

No, $T(\mathbf{x}) \neq B\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^2$. For example,

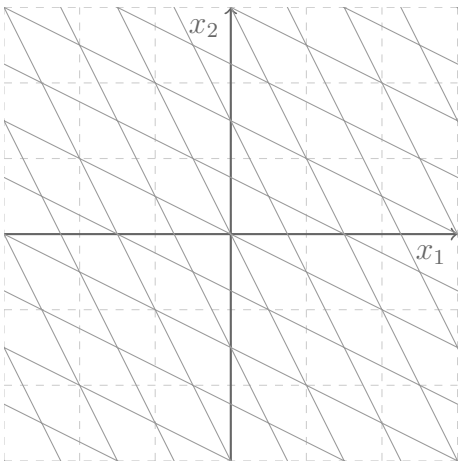
$$\begin{bmatrix}5\\0\\1\end{bmatrix} = T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) \neq \begin{bmatrix}1 & 0\\0 & 1\\0 & 0\end{bmatrix}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\-1\\0\end{bmatrix}.$$

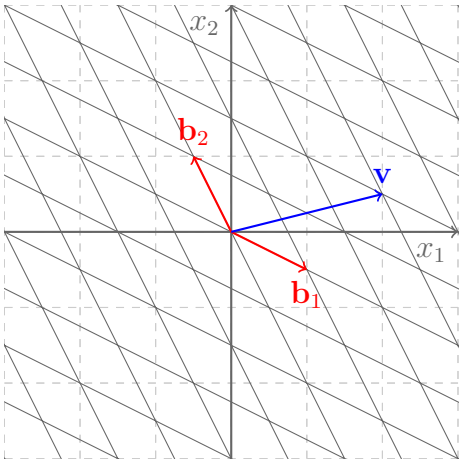
(5) Is $T(\mathbf{v})_{\mathcal{C}_2} = T_{\mathcal{C}_2,\mathcal{C}_1}\mathbf{v}_{\mathcal{C}_1}$ for all $\mathbf{v} \in \mathbb{R}^2$?

Yes. See Theorem 2 in Lecture 16.

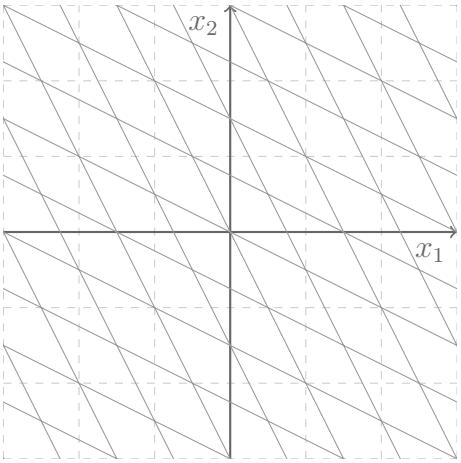
11. *Consider the following basis* $\mathcal{B} := \left(\mathbf{b}_1 = \begin{bmatrix}1\\-0.5\end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix}-0.5\\1\end{bmatrix}\right)$ *of* $\mathbb{R}^2$. *Let* $\mathbf{v} = 3\mathbf{b}_1 + 2\mathbf{b}_2$.

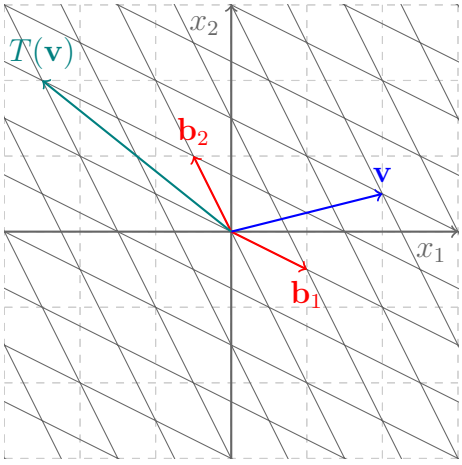
(i) *Draw* $\mathbf{b_1}, \mathbf{b_2}$ *and* $\mathbf{v}$ *in the graph below.*





(ii) *Consider a linear transformation* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *such that* $T_{\mathcal{BB}} = \begin{bmatrix} -\frac{2}{3} & 0 \\ 0 & 0.5 \end{bmatrix}$. *Draw* $T(\mathbf{v})$.





12. *For each of the following functions, determine if they are linear transformations.*

(1) $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$

This is not a linear transformation. Scalar multiplication is violated, for example:

$$(-1)T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \neq T\left((-1)\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(2) $\det : M_{2\times 2} \to \mathbb{R}$ such that $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$, where $M_{2\times 2}$ is the vector space of all 2×2 matrices.

This is not a linear transformation. Scalar multiplication is violated, for example:

$$\det\left(2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 4 \neq 2\det\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 2$$

(3) $\mathrm{tr} : M_{2\times 2} \to \mathbb{R}$ such that $\mathrm{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d$, where $M_{2\times 2}$ is the vector space of all 2×2 matrices.

This is a linear transformation.

(4) $T : M_{3\times 3} \to M_{3\times 3}$ such that $T\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & a & b \\ 0 & d & e \end{bmatrix}$ where $M_{3\times 3}$ is the vector space of all 3×3 matrices.

This is a linear transformation.

13. *True or False? Justify your answers.*

(1) Let $T : V \to W$ be a linear transformation and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in V. If $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ are linearly independent, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are also linearly independent.

True, since if $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + ... + x_n\mathbf{v}_n = 0$ then $x_1T(\mathbf{v}_1) + x_2T(\mathbf{v}_2) + ... + x_nT(\mathbf{v}_n) = 0$, but $T(\mathbf{v}_1), T(\mathbf{v}_2), ..., T(\mathbf{v}_n)$ are linearly independent so all x_is are equal to 0.

(2) Let $T : V \to W$ be a linear transformation and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in V. If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, then $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ are also linearly independent.

False, consider $T : \mathbb{R}^3 \to \mathbb{R}$ such that $T(\mathbf{v}) = 0$.

(3) Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V and let $T : V \to W$. Do the images $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ span W?

This is false. To see this map the vectors $\mathbf{v}_i$ to zero. The images $T\mathbf{v}_i$ with $i = 1, 2, \cdots, n$ will not span W.

14. *This problem explores the connection between linear algebra and quantum mechanics. In particular, we interpret in a particular case, the so called* ***Heisenberg's uncertainty principle*** *which can be stated as follows: It is impossible to have absolute precision while measuring simultaneously the position and momentum of a quantum particle. In terms of linear transformations, this principle asserts that the linear transformations associated to position and momentum do not commute.*

(1) **(The position operator).**Consider the case in which the spaces of quantum states are vector spaces of polynomials P_k (of degree at most k). Consider the linear transformation $\hat{x} : P_{n-1} \to P_n$ defined as

$$\begin{aligned} \hat{x} : P_{n-1} &\to P_n \\ p(t) &\mapsto tp(t). \end{aligned}$$

For instance $\hat{x}(t^2-t+2) = t^3-t^2+2t$. Check that this is indeed a linear transformation and find its matrix representation $\hat{X}$ with respect to the standard basis of P_{n-1} and P_n.

Let $p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_{n-1}t^{n-1}$ and $q(t) = b_0 + b_1t + b_2t^2 + \cdots + b_{n-1}t^{n-1}$ be two polynomials of degree at most $n-1$. In order to check that $\hat{x}$ is a linear transformation, we verify that the transformation respects linear combinations:

$$\begin{aligned} \hat{x}(cp(t) + dq(t)) &= t(cp(t) + d(q(t))) \\ &= t(c(a_0 + a_1t + a_2t^2 + \cdots + a_{n-1}t^{n-1}) + d(b_0 + b_1t + b_2t^2 + \cdots + b_{n-1}t^{n-1})) \\ &= c(a_0t + a_1t^2 + a_2t^3 + \cdots + a_{n-1}t^n) + d(b_0t + b_1t^2 + b_2t^3 + \cdots + b_{n-1}t^n) \\ &= c\hat{x}(p(t)) + d\hat{x}(q(t)). \end{aligned}$$

With respect to the standard bases of P_{n-1} and P_n we get that:

$$\begin{aligned} \hat{x}(1) &= t \\ \hat{x}(t) &= t^2 \\ &\vdots \\ \hat{x}(t^{n-1}) &= t^n, \end{aligned}$$

therefore $\hat{X} = [\hat{x}(1), \hat{x}(t), \hat{x}(t^2), \cdots, \hat{x}(t^{n-1})]$ is the $(n+1 \times n)$- matrix

$$\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & 0 \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

(2) **(The momentum operator).** Now consider the transformation $\hat{q} : P_n \to P_{n-1}$ defined as

$$\begin{aligned} \hat{q} : P_n &\to P_{n-1} \\ p(t) &\mapsto \frac{d}{dt}p(t). \end{aligned}$$

For instance $\hat{q}(t^3-t^2+2) = 3t^2-2t$. Check that this is indeed a linear transformation and find its matrix representation $\hat{Q}$ with respect to the standard basis of P_n and P_{n-1}.

The linearity of the transformation $\hat{q}$ was checked in Example 5, Lecture 16. With respect to the standard bases of P_n and P_{n-1} we get that:

$$\begin{aligned} \hat{q}(1) &= 0 \\ \hat{q}(t) &= 1 \\ \hat{q}(t^2) &= 2t \\ &\vdots \\ \hat{q}(t^{n-1}) &= (n-1)t^{n-2} \\ \hat{q}(t^n) &= nt^{n-1}, \end{aligned}$$

therefore $\hat{Q} = [\hat{q}(1), \hat{q}(t), \hat{q}(t^2), \cdots, \hat{q}(t^n)]$ is the $(n \times n+1)$- matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 3 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 & \\ \vdots & & & & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & n \end{bmatrix}.$$

(3) **(The uncertainty principle).**Let $p(t)$ be a polynomial. Check that

$$\hat{x}\hat{q}(p(t)) \neq \hat{q}\hat{x}(p(t)).$$

This means that the position and momentum operators do not commute!

Let $p(t) = a_0 + a_1 t + a_2 t^2 + \cdots a_n t^n$. Then

$$\begin{aligned} \hat{x}\hat{q}(p(t)) &= \hat{x}(\hat{q}(a_0 + a_1 t + a_2 t^2 + \cdots a_n t^n)) \\ &= \hat{x}(a_1 + 2a_2 + 3a_3 t^2 + \cdots + n a_n t^{n-1}) \\ &= a_1 t + 2a_2 t^2 + 3a_3 t^3 + \cdots n a_n t^n. \end{aligned}$$

In the other hand

$$\begin{aligned} \hat{q}\hat{x}(p(t)) &= \hat{q}(\hat{x}(a_0 + a_1 t + a_2 t^2 + \cdots a_n t^n)) \\ &= \hat{q}(a_0 t + a_1 t^2 + a_2 t^3 + \cdots + a_n t^{n+1}) \\ &= a_0 + 2a_1 t + 3a_2 t^2 + \cdots (n+1) a_n t^n. \end{aligned}$$

Clearly $\hat{x}\hat{q}(p(t)) \neq \hat{q}\hat{x}(p(t))$!

In terms of the matrix representations $\hat{X}$ and $\hat{Q}$ we obtain that $\hat{Q}\hat{X}$ is the $(n \times n)$- diagonal matrix:

$$\hat{Q}\hat{X} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 2 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 3 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & & k \cdots & 0 \\ \vdots & & & & \ddots & \\ 0 & 0 & 0 & 0 & \cdots & n \end{bmatrix}.$$

whereas $\hat{X}\hat{Q}$ is the $(n+1 \times n+1)$- diagonal matrix:

$$\hat{X}\hat{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & & k \cdots & 0 \\ \vdots & & & & \ddots & \\ 0 & 0 & 0 & 0 & \cdots & n \end{bmatrix}.$$

These two matrices are not the same, in fact they do have different sizes!

(4) **(The Heisenberg relation).** Now consider that the space of quantum states is the space $F(\mathbb{R})$ of real functions. The position and momentum operators are defined as above. The commutator $[\hat{q}, \hat{x}]$ of $\hat{q}$ and $\hat{x}$ is defined to be the linear transformation $\hat{q}\hat{x} - \hat{x}\hat{q} : F(\mathbb{R}) \to F(\mathbb{R})$. Use your knowledge of calculus to check that, for every function f:

$$[\hat{q}, \hat{x}]f = f.$$

This is another instance of the Heisenberg uncertainty principle!

We will use the product rule for the derivative:

$$\frac{d}{dt}(fg(t)) = f\frac{d}{dt}g(t) + g\frac{d}{dt}f(t).$$

Now, we can compute:

$$\hat{q}\hat{x}(f(t)) = \frac{d}{dt}(tf(t)) = f(t) + t\frac{d}{dt}f(t).$$

In the other hand,

$$\hat{x}\hat{q}(f(t)) = t\frac{d}{dt}f(t).$$

Therefore:

$$[\hat{q}, \hat{x}]f(t) = \hat{q}\hat{x}(f(t)) - \hat{x}\hat{q}(f(t)) = f(t) + t\frac{d}{dt}f(t) - t\frac{d}{dt}f(t) = f(t),$$

as we wanted.